

Final Assessment: OTW Krypton Challenge

BACKGROUND:

The goal of this assignment is to challenge yourself with another CTF. In this assignment, you will work on the Over the Wire Challenge called Krypton. Krypton is a "wargame" aimed at novices. The idea is to complete each level by achieving the objective listed in the challenge. Please read the instructions below for help with starting and completing the challenge.

INSTRUCTIONS:

Log into the challenge using the following SSH command:

```
ssh krypton1@krypton.labs.overthewire.org -p 2231
```

CD into the krypton/ directory to find the challenges

Since this is the first challenge, you will CD into the krypton1 folder (you will do the same for each challenge, incrementing the number)

Attempt to read the file. Depending on the challenge, you may have to use a tool to decrypt the password.

Once you have the password, type exit and then log back in using the next level (for example, the next level in this example would be krypton2)

KRYPTON LEVEL 0 → LEVEL 1:

The screenshot shows the CyberChef web application interface. The browser address bar displays the URL: `gchq.github.io/CyberChef/#recipe=From_Base64('A-Za-z0-9%2B/%3D',true,false)&input=CiMxSlpVRlJQVGtsVFhSkZRVE9Cg`. The interface is divided into three main sections: **Operations**, **Recipe**, and **Input/Output**.

- Operations:** A sidebar on the left containing various tools like "To Base64", "From Base64", "To Hex", "From Hex", "To Hexdump", "From Hexdump", "URL Decode", "Regular expression", "Entropy", "Fork", "Magic", "Data format", "Encryption / Encoding", "Public Key", and "Arithmetic / Logic".
- Recipe:** The central area shows a selected recipe "From Base64" with a dropdown menu set to "Alphabet" and "A~Za~z0~9+/= ". There are checkboxes for "Remove non-alphabet chars" (checked) and "Strict mode" (unchecked).
- Input:** The top right section contains the input text: `S1JZUFRTk\LTR1JFQVQ=`.
- Output:** The bottom right section shows the result of the conversion: `KRYPTONISGREAT`.

At the bottom of the interface, there is a "STEP" indicator, a green "BAKE!" button, and an "Auto Bake" checkbox which is checked.

Rationale: I used Cyberchef to translate data format from Base64 to plain text.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 1 → LEVEL 2:

The screenshot shows an AWS CloudShell terminal window. The terminal prompt is `krypton1@krypton:/krypton/krypton1$`. The user has entered the following commands in order:

```
1 ls
2 cd ..
3 ls
4 cd krypton2
5 ls
6 ls -al
7 cd ..
8 ls
9 cd krypton
10 cd krypton1
11 ls
12 ls -al
13 pwd
14 /home/krypton1
15 cd ..
16 clear
17 clear
18 cd -
19 ls
20 ls -al
21 cd /krypton
22 ls
23 cd /krypton1
24 cd krypton1
25 ls
26 cat README
27 LS
28 ls
29 cat krypton2
30 clear
31 history
```

The output of the `history` command is displayed at the bottom of the terminal. The terminal window also shows the AWS IAM console navigation bar at the top and the CloudShell interface at the bottom.

The screenshot shows the CyberChef web application interface. The URL bar displays the recipe `gchq.github.io/CyberChef/#recipe=ROT13(true,true,false,13)&input=WVJJUikgR0pCIENORkZKQkVRIEVCR0dSQQ`. The left sidebar lists various recipes, with `ROT13` selected. The main area shows the `ROT13` recipe configuration with the following settings:

- ☒ Rotate lower case chars
- ☒ Rotate upper case chars
- ☐ Rotate numbers
- Amount: 13

The input field contains the text `YRIRY GJB CNFFJBEQ EBGGRRA`. The output field displays the result: `LEVEL TWO PASSWORD ROTTEN`. The interface also includes a 'BAKE!' button and a 'Raw Bytes' view option.

Final Assessment: OTW Krypton Challenge

Rationale: Most Krypton levels are stored in /krypton/, so start by going there. The README tells me the level's message is encrypted with ROT13, which is just a Caesar cipher. I used the ROT13 decoder via CyberChef to receive the password **ROTTEN**.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 2 → LEVEL 3:

```
chrome
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-06f...
krypton2@krypton:/tmp/tmp.GV0hpj8MDs$ history
1 od /krypton
2 ls
3 cd krypton2
4 ls
5 cat README
6 ls
7 od encrypt
8 od /encrypt
9 mktemp -d /tmp/tmp.J2V2025Krp
10 mktemp -d /tmp/tmp.J2V2025Kry
11 cd -
12 od -
13 mktemp -d /tmp/tmp.WE20nCpCQ
14 od /tmp
15 clear
16 mktemp -d
17 /tmp/tmp.WE20nCpCQ
18 mktemp -d
19 /tmp/tmp.WE20nCpCQ
20 ls
21 clear
22 ls -al
23 cd /krypton
24 ls
25 cd krypton2
26 ls
27 cat keyfile.dat
28 cat keyfile.dat
29 ls -l
30 cat README
31 ls
32 ls
33 cat krypton3
34 mktemp -d /tmp/tmp.J20i0dhtb
35 mktemp -d /tmp/tmp.J20kwdjhf
36 mktemp -d
37 od /tmp/tmp.OV0hpj8MDs
38 ln -s /krypton/krypton2/keyfile.dat
39 ls
40 chmod 777 .
41 /krypton/krypton2/encrypt /etc/issue
42 ls
43 cat /etc/issue
44 cat ciphertext
45 cat /krypton/krypton2/krypton3
46 clear
47 history
krypton2@krypton:/tmp/tmp.GV0hpj8MDs$
```

i-06f1bd3f330c4c2e0 (HOMEWORK)

cryptii Visibility matters

VIEW
Text
OMQEMDUEQMEK

ENCODE DECODE
Caesar cipher
SHIFT
- 12 a→m +
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
→ Decoded 12 chars

VIEW
Text
CAESARISEASY

Caesar cipher: Encode and decode online

Method in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet. The method is named after Julius Caesar, who used it in his private correspondence.

Decimal to text URL encode Enigma decoder Text to binary Commercial Enigma

Open in
ciphereditor

Final Assessment: OTW Krypton Challenge

Rationale: For Level 2 to 3, I started by navigating to the challenge directory with `cd /krypton` and then into `krypton2` to read the `README` file. The instructions explained that this level uses a Caesar cipher, so I needed to figure out the shift. I created a temporary folder with `mktemp -d` and moved into it, then made a symbolic link to `keyfile.dat` using `ln -s /krypton/krypton2/keyfile.dat` and set permissions with `chmod 777`. Finally, I used Cryptii to decode the encrypted key with this shift, which gave me the password for the next level: **CAESARISEASY**.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 3 → LEVEL 4:

```
Chrome File Edit View History Bookmarks Profiles Tab Window Help
OverTheWire x Final Assess x Instances | E x EC2 Instance x Frequency x Online Lett x OverTheWire x quipquip - c x + Gemini
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-06f1bd3f330c4c...
IAM Elastic Container Service EC2 Billing and Cost Manage... CloudShell EFS

krypton3@krypton:/krypton/krypton3$ history
1 cd /krypton/krypton3
2 ls
3 cat README
4 cat HINT1
5 cat HINT2
6 cat found1
7 cat found2
8 cat found3
9 cat krypton4
10 clear
11 history
krypton3@krypton:/krypton/krypton3$
```

i-06f1bd3f330c4c2e0 (HOMEWORK)

PublicIP: 3.236.4.153 PrivateIP: 172.31.68.217

CloudShell Feedback Console Mobile App

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OverTheWire x Final Assess x Instances | E x EC2 Instance x Frequency x Online Lett x OverTheWire x quipquip - c x + Gemini

md5decrypt.net/en/Letters-frequency-analysis/?ref=learnhacking.io

Encrypt / Decrypt
Obfuscation
Ciphers
Conversion
Network
API
Contact
FR / EN

English letters frequencies

Frequency Analysis

3-Month Unlimited Plan \$15/mo.

A Black Friday deal you don't want to miss—switch to Mint and save big on your phone bill.

Mint Mobile

Fréquence des lettres du Chiffre

Lettr	Pourcentage d'apparition
s	12%
q	10%
j	8%
g	7%
c	6%
n	5%
b	4%
u	3%
d	2%
v	1%
w	1%
z	1%
e	1%
k	1%
m	1%
a	1%
x	1%
y	1%
i	1%
t	1%
f	1%
l	1%
o	1%
p	1%
r	1%

Fréquence des lettres de la langue sélectionnée

Lettr	Pourcentage d'apparition
e	12%
t	10%
a	8%
o	7%
i	6%
n	5%
s	4%
h	3%
r	2%
d	1%
l	1%
c	1%
u	1%
m	1%
w	1%
f	1%
g	1%
y	1%
p	1%
b	1%
v	1%
k	1%
j	1%
x	1%
q	1%
z	1%

Privacy - Terms

Final Assessment: OTW Krypton Challenge

Options

N-gram size: 3

☐ Preserve casing
☐ Keep spaces & non-letters
☐ Remove accents

Step size: 1

Offset: 0

Results

Code	Count	Frequency
JDS	19	1.481 %
DSN	11	0.857 %
QGW	11	0.857 %
SUQ	10	0.779 %
JCB	10	0.779 %
CBG	10	0.779 %

Comparison

Language: English

Code	Frequency
THE	1.897 %
AND	0.944 %
ING	0.762 %
HER	0.584 %

Clues: For example G-R-I-Q-W-W-T-H-E

Solve

New Frames Have Arrived

automatically selected patristical mode; you can override by using the drop down menu next to the solve button.

in cryptography a caesar cipher also known as a caesars cipher the shift cipher caesars code or caesar shift is one of the simplest and most widely known encryption techniques it is a type of substitution cipher in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet for example with a shift of a would be replaced by b would become e and soon the method is named after julius caesar who used it to communicate with his generals the encryption step performed by a caesar cipher is often incorporated as part of more complex schemes such as the vigenre cipher and still has modern application in the rot system as with all single alphabet substitution ciphers the caesar cipher is easily broken and in practice offers essentially no communication security shakespeare produced most of his known work between and his early plays were mainly comedies and histories genre he raised to the peak of sophistication and artistry by the end of the sixteenth century next he wrote mainly tragedies until about including hamlet king lear and macbeth considered some of the finest examples in the english language in his last phase he wrote tragicomedies also known as romances and collaborated with other playwrights many of his plays were published in editions of varying quality and accuracy during his lifetime and in two of his former theatrical colleagues published the first folio a collected edition of his dramatic works that included all but two of the plays now recognised as shakespeares well done the level four password is brute

in krzt so greth zeke a mer kith a red nox now nemeke a merm it harsham h if skit hark eamer m koya or keamer m h if simona of shani ltd a m se n y loms wiya dzx now nan krztstion saknhi quae is i mesta of mucus is usion kithar in whikha ekh dassar in shat dein sab sinrot deokay czed as sarm olaf ibay nul car of to mision my own shaed the cas for a beltd a wish emh if so few oudy car at deokay czyc woudy cakola aeny moon shalas hoy im ne laye fsar judi unkeamer who umay is sokol lunikes a wish him gana red m shaan krztstion ns at tar for layczeke a mer kith a rim of sanik or to resay enters of lora kolt dab mkhalam mukhem shap' deed the casm ucns is usion kit harm shake a mer kit harima em

with are nym siddhem loy arnett dikesion in sharos mzm sakem wished d ming iane mv in trek sika of farm amman sied dzno kolluni kesion nakuri szm i ramhare i may so shata ex of iusin k duyng held asking regikolay i an edmox now irzing que diszen yekkurek i yayision of himyre lesikworxm i crus a

Rationale: For Level 3 to 4, I started by navigating to the challenge directory with `cd /krypton/krypton4`. I first read the *HINT* file to understand the challenge, then examined the ciphertext files with `cat`

Final Assessment: OTW Krypton Challenge

found1 and *cat found2*. To decode the messages, I used a combination of online tools: QuipQuip decoder, Boxentriq frequency analysis, and MD5Decrypt letter frequency analysis. By analyzing the letter frequencies and applying substitutions suggested by these tools, I was able to reconstruct the plaintext and move on to the next level using password **BRUTE**.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 4 → LEVEL 5:

The screenshot shows the AWS Management Console interface. The top navigation bar includes links for IAM, Elastic Container Service, EC2, Billing and Cost Management, CloudShell, and EFS. The main content area displays a terminal window for an EC2 instance. The terminal output shows the following commands and results:

```
krypton4@krypton:/krypton/krypton4$ history
1 clear
2 cd /krypton/krypton4
3 ls
4 cat HINT
5 cat found1
6 cat found2
7 cat krypton5
8 clear
9 history
krypton4@krypton:/krypton/krypton4$
```

Below the terminal window, the instance ID **i-06f1bd3f330c4c2e0 (HOMEWORK)** is displayed, along with its public and private IP addresses: **Public IP: 3.236.4.153** and **Private IP: 172.31.68.217**. The bottom of the console shows the CloudShell logo, a feedback link, a console mobile app link, and the Amazon Web Services copyright notice.

The screenshot shows the **dcode.fr/vigenere-cipher** website. The page is divided into three main sections: a list of ciphertexts on the left, a central decryption tool, and a list of similar pages on the right. The central tool is set to "DECRYPTION METHOD" and "KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6". The "FREKEY" ciphertext is selected, and the "FREKEY" key is entered. The "DECRYPT" button is clicked, and the resulting plaintext is displayed in the "VIGENERE PLAIN TEXT" field.

The list of ciphertexts on the left includes:

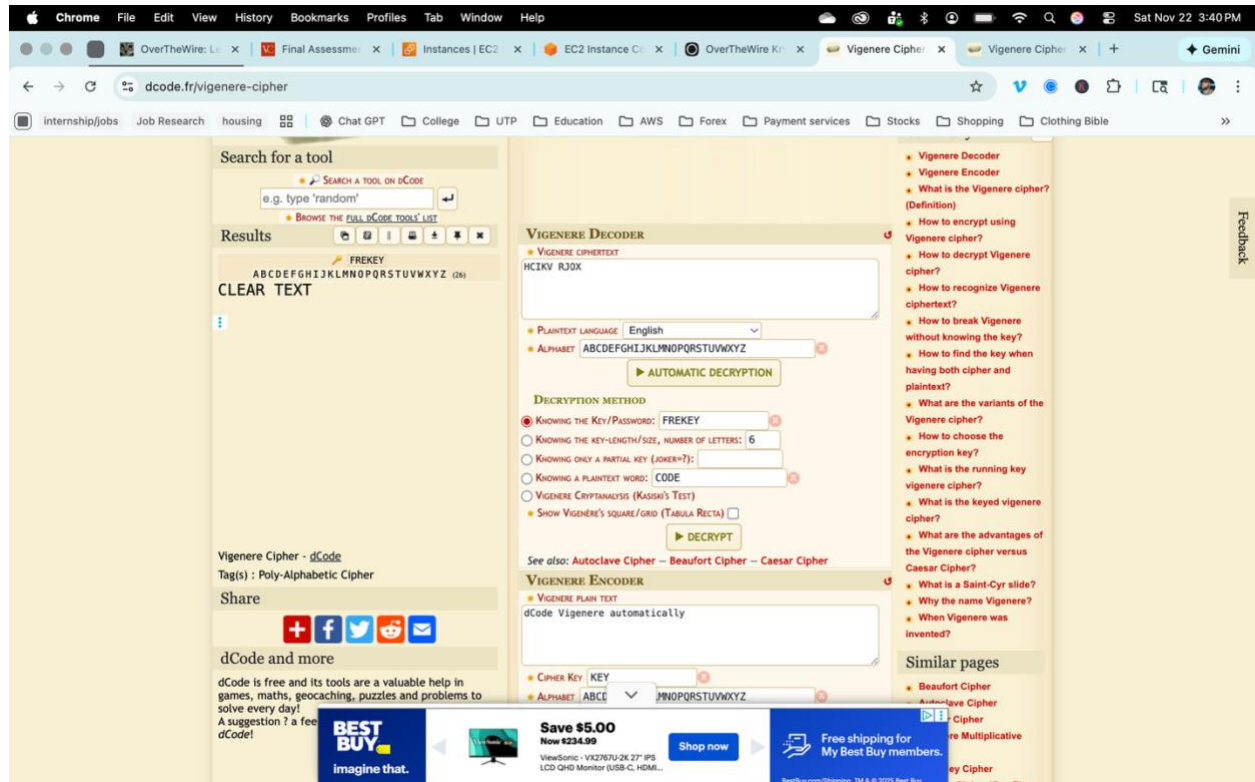
- IRCTH ENLKE WJCHN WJRMN ENCTH
- EGUAR DIANO FTHEG ATESL IVEOT
- HISOF FICER UNLOC KEOTH EIRSP
- ECTAC LESTO PUTTH EMBAC KINHI
- SGREA TBOXA NOTHE NHEPO LITEL
- YOPEN EOTHE GATEF OROUR FRIEN
- DSMHI CHROA DLEAD STOTH EWICK
- EDMIT CHOFT HEWES TASKE DDORO
- THYTH EREIS NOROA DANSW EREDT
- HEGUA RDIAN OFTHE GATES NOONE
- EVERW ISHES TOGOT HATMA YHOWT
- HENAR EMETO FINDH ERINO UIREO
- THEGI RLTHA TWILL BEEAS YREPL
- IEDTH EMANF ORWHE NSHEK NOMSY
- QUARE INTHE COUNT RYOFT HEWIN
- KIESS HEWIL LFIND YOAN DMAKE
- YOUAL LHESR LAVES PERHA PSNOT
- SAIDT HESCA REORO WFORW MEAN
- TODES TROYH EROHT HATIS DIFFE
- RENTS AIDTH EGUAR DIANO FTHEG
- ATESN OONEH ASEVE RDEST ROYED
- HERBE FORES OINAT URALL YTHOU
- GHTSH ENOUL DMAKE SLAVE SOFYO
- UASSH EHASO FTHER ESTBU TTAKE
- CAREF ORSHE ISWIC KEDAN DFIER
- CEAND MAYNO TALLD WYOUT ODEST
- ROYHE RKPEE TOTHE WESTW HERET
- HESUN SETSA NOYOU CANNO TFAIL
- TOFIN DHERT HEYTH ANKED HIMAN
- DBADE HINGO ODBYE ANOTU RNEDT
- OWARD THEWE STWAL KINGO VERFI
- ELDOS FSOFT GRASS DOTTE DHERE
- ANDTH EREWI THDAI SIESA NDBUT
- TERCU PSDOR OTHYS TILLW ORETH
- EPRET TYSIL KDRES SSHEH ADPUT
- ONINT HEPAL ACEBU TNOWT OHERS
- URPRI SESHE FOUNO ITMAS NOLON
- GERE
- BBON
- OLOE
- SWHI
- MERJ

The central decryption tool shows the following fields and buttons:

- PLAINTEXT LANGUAGE:** English
- ALPHABET:** ABCDEFGHIJKLMNOPQRSTUVWXYZ
- AUTOMATIC DECRYPTION:** [Button]
- DECRYPTION METHOD:** [Radio buttons]
- KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS:** 6
- KNOWING ONLY A PARTIAL KEY (JOKER=I):** FREKEY
- KNOWING A PLAINTEXT WORD:** CODE
- VIGENERE CRYPTANALYSIS (KASISKI'S TEST):** [Radio button]
- SHOW VIGENERE'S SQUARE/GRID (TABULA RECTA):** [Radio button]
- DECRYPT:** [Button]
- VIGENERE ENCODER:** [Section header]
- VIGENERE PLAIN TEXT:** dCode Vigenere automatically
- CIPHER KEY:** KEY
- ALPHABET:** ABCDEFGHIJKLMNOPQRSTUVWXYZ
- PRESERVE PUNCTUATION, LOWERCASE ETC.:** [Checked]
- SHOW VIGENERE'S SQUARE/GRID (TABULA RECTA):** [Radio button]
- ENCRYPT:** [Button]

The right sidebar contains a "Feedback" button, a "Similar pages" section with links to Beaufort Cipher, Autoclave Cipher, Caesar Cipher, Vigenere Multiplicative Cipher, Vigenere Cipher, and Vernam Cipher (One Time Pad), and a "Support dCode" section with a link to the Discord forum.

Final Assessment: OTW Krypton Challenge



Rationale: For Level 4 to 5, I started by using *cd /krypton/krypton4*. I read the *HINT* file and then viewed the ciphertext files using *cat found1* and *cat found2*. After checking *krypton5* to confirm the next step, I analyzed the ciphertexts using Dcode's Vigenère cipher tool, since the *README* explained this level used a Vigenère cipher with a key length of 6. By entering the text and the key length into Dcode, I was able to determine the correct key. Using that key, I decrypted the short challenge ciphertext and obtained the final answer: **CLEARTEXT**.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 5 → LEVEL 6:

Chrome | File | Edit | View | History | Bookmarks | Profiles | Tab | Window | Help

OverTheWire: Level Info: Kryp... | Instances | EC2 | us-east-1 | EC2 Instance Connect | us-east-1 | OverTheWire Krypton (Level 5) | Vigenere Cipher Solver - Online | Gemini

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-06f1bd3f33...

Internship/jobs | Job Research | housing | Chat GPT | College | UTP | Education | AWS | Forex | Payment services | Stocks | Shopping | Clothing Bible

IAM | Elastic Container Service | EC2 | Billing and Cost Manage... | CloudShell | EFS

```
krypton5@krypton:/krypton/krypton5$ history
1 clear
2 cd /krypton/krypton5
3 ls
4 cat README
5 cat found1
6 cat found2
7 cat found3
8 cat krypton6
9 clear
10 history
krypton5@krypton:/krypton/krypton5$
```

i-06f1bd3f330c4c2e0 (HOMEWORK)

Public IP: 3.216.4.153 | Private IP: 172.31.68.217

CloudShell | Feedback | Console Mobile App

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Chrome | File | Edit | View | History | Bookmarks | Profiles | Tab | Window | Help

OverTheWire: Level Info: Kryp... | Instances | EC2 | us-east-1 | EC2 Instance Connect | us-east-1 | OverTheWire Krypton (Level 5) | Vigenere Cipher Solver - Online | Gemini

dcode.fr/vigenere-cipher

Internship/jobs | Job Research | housing | Chat GPT | College | UTP | Education | AWS | Forex | Payment services | Stocks | Shopping | Clothing Bible

Warning: Showing most likely results

Search for a tool

SEARCH A TOOL ON DCODE

e.g. type "random"

BROWSE THE FULL DCODE TOOLS LIST

Results

ABCDEFGHIJKLMNOPQRSTUVWXYZ (26)

14

12

NPYPPO GBLUJ OVDBG CTXXE VLFNF
OXBKT HPBZA BRAUX PJUZH MACVA
BZDBX QZBEG QHLYD GPCR1 MYTKN
XOUHL LXEVN WPMOX JLOPD BZIXQ
EGUEG QHLYD GTLBN OHKEP RMYXB
ENPAX PJUZH MAUTW FIUHK HKVDG
CAPFO VNAFY HLTJQ UZBLR GJOUX
PJUZH MAUEN VPMKH DZNAV NDTXP
JTSYH AXMKH SAFJH BWSGW WQYCX
WUWNE AAXKX KTTOF OLAZF PDPUC
UZGAH DNOZM PJSGG ANFSZ TEPVZ
VLLVY POMAN PLGBA YTNRU SELKX
CCQCY XHPVW ABNOX WSCNE AMOHV
PVMAC YBTYZ YPOFI MTWKK ZAGBL
IWAUT JGJLK NKFTD NMZHR AQUEG
MUHNO KTOGU BMKKT XPVYZ BSOXH
PRXVG NPUQA VNMKJ YGXRR XMHMC
QDZBL MHMAG XHVHA AJOME AEFHM
NRGSA TLLTX BEDIC UYBKK PDJYS
AAJNG LAEYH DNJIL EGBHE FNITF
NQHGI WSJAR HDBYD CEHNC UYFYG
DJGBL MMNOQ ABZLG LHCCZ GBLKJ
SGGAN EPLGS KIDNF HKLAL PSNKK
TVOGT JJCAA FBCXN SUJXT JMWAG
BYHSA QUBEU UVJEP QKGBJ HZJVG

KEYLENGTH

dCode is preparing a new interface. Come test and give your feedback on the new page: **Vigenere Cipher!**

VIGENERE CIPHER

Cryptography - Poly-Alphabetic Cipher - Vigenere Cipher

VIGENERE DECRYPTER

VIGENERE CIPHERTEXT

QOIVY XURRY BHCSS SKOLX XOSEQ RTAOP WNSZK MVDLC PRTRB ZRGPZ
AAGGK ZIMAP RLKVV EAZRT XJZCS DRIVVZ BZRWK NNRIM ZSRYX IEDVH
GLQHL FZOHK KEESK KEHDI PLZRV KVFIB XSEKB TZSPE EADNV DLSY
ZSGYK GCELN TTUIG NQHT BJKKG ZRFEK ABIAF MIKWA RVNFK UGGFY
JRSIP NBUIJ LDDSZ ALMSA VPMTX IBSDM

PLAINTEXT LANGUAGE: English

ALPHABET: ABCDEFGHIJKLMNOPQRSTUVWXYZ

AUTOMATIC DECRYPTION

DECRYPTION METHOD

☒ KNOWING THE KEY/PASSWORD: KEY

☐ KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6

☐ KNOWING ONLY A PARTIAL KEY (JOKER-?): FREKEY

☐ KNOWING A PLAINTEXT WORD: CODE

☐ VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

SHOW VIGENERE'S SQUARE/GRID (TABULA RECTA)

DECRYPT

See also: Autoclave Cipher - Beaufort Cipher - Caesar Cipher

VIGENERE ENCODER

VIGENERE PLAIN TEXT

dCode Vigenere automatically

CIPHER KEY: KEY

ALPHABET: ABCDEFGHIJKLMNOPQRSTUVWXYZ

PRESERVE PUNCTUATION, LOWERCASE ETC. ☒

SHOW VIGENERE'S (TABULA RECTA)

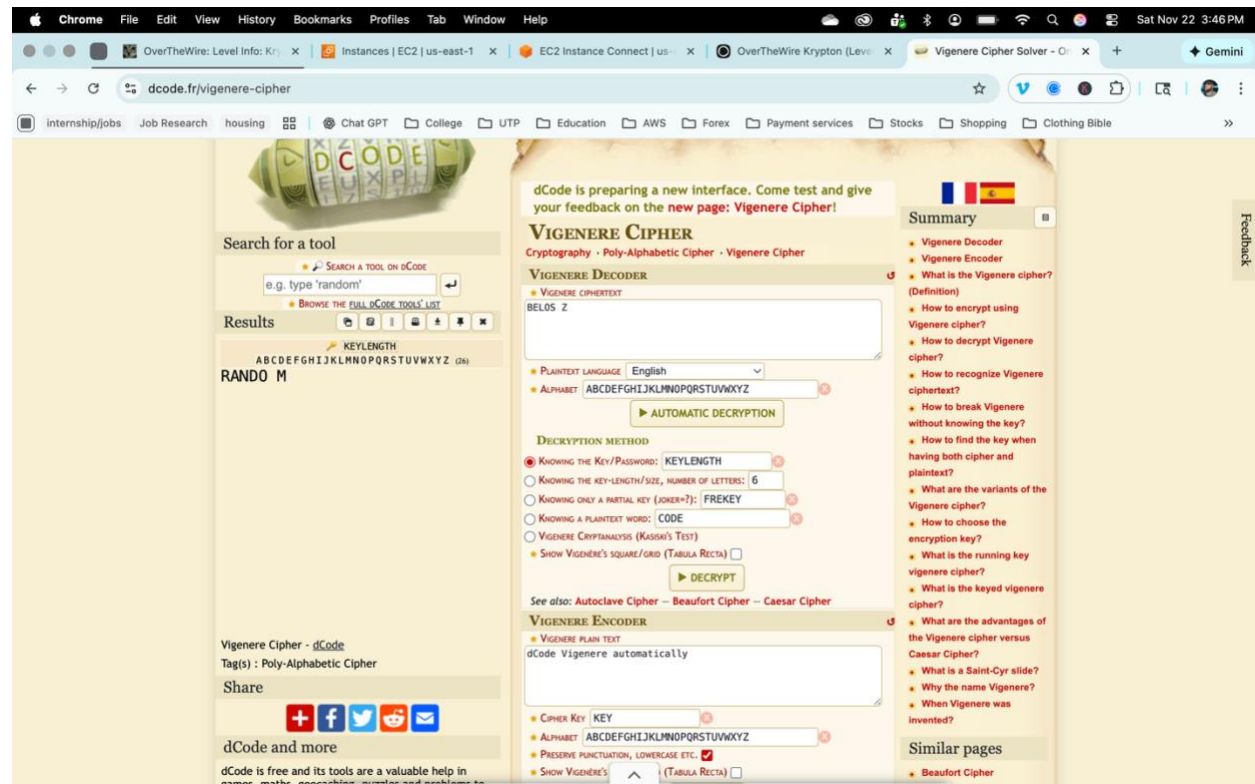
Summary

- Vigenere Decoder
- Vigenere Encoder
- What is the Vigenere cipher? (Definition)
- How to encrypt using Vigenere cipher?
- How to decrypt Vigenere cipher?
- How to recognize Vigenere ciphertext?
- How to break Vigenere without knowing the key?
- How to find the key when having both cipher and plaintext?
- What are the variants of the Vigenere cipher?
- How to choose the encryption key?
- What is the running key vigenere cipher?
- What is the keyed vigenere cipher?
- What are the advantages of the Vigenere cipher versus Caesar Cipher?
- What is a Saint-Cyr slide?
- Why the name Vigenere?
- When Vigenere was invented?

Similar pages

- Beaufort Cipher

Final Assessment: OTW Krypton Challenge



Rationale: For Level 5 to 6, I started by using `cd /krypton/krypton5`. I read the **README** first, then viewed all three ciphertext files using `cat found1`, `cat found2`, and `cat found3` to see what I was working with. After that, I checked the next-level file, `krypton6`, to gain additional information. Since the **README** mentioned that no key length would be provided, I copied the text from `found1` and used Dcode's Vigenère Automatic Decryption tool. Dcode handled the analysis for me—it detected both the key length and the key itself. The key turned out to be **KEYLENGTH** and using it to decrypt the challenge ciphertext granted me the key for the next level which is **RANDOM**.

Final Assessment: OTW Krypton Challenge

KRYPTON LEVEL 6 → LEVEL 7:

```
Chrome File Edit View History Bookmarks Profiles Tab Window Help
OverTheWire: Level 6 | us-east-1 | EC2 Instance Connect | Krypton_Scripts/fre... | No content provided | Vigenere Cipher Sol... | Gemini
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-06f1bd3f33...
IAM Elastic Container Service EC2 Billing and Cost Manage... CloudShell EPS
hisotyr: command not found
krypton6@krypton:/tmp/tmp.75Shlmvg02$ history
1 cd /krypton/krypton6
2 cat README
3 clear
4 cat HINT1
5 cat HINT2
6 cat krypton7
7 mktemp -d
8 ls
9 cd /tmp/tmp.75Shlmvg02
10 ln -s /krypton/krypton6/keyfile.dat
11 ls
12 chmod 777 .
13 touch tale.txt
14 ls
15 name tale.txt
16 nano
17 clear
18 ls
19 cat tale.txt
20 /krypton/krypton6/encrypt6 tale.txt ciphertale
21 ls
22 cat ciphertale
23 xxd -b tale.txt
24 xxd -b ciphertale
25 python3 -c "print('A' * 100)" > test.txt
26 ls
27 cat test.txt
28 /krypton/krypton6/encrypt6 test.txt cipher_t.txt
29 cat cipher_t.txt
30 cat /krypton/krypton6/krypton7
31 clear
32 hisotyr
33 history
krypton6@krypton:/tmp/tmp.75Shlmvg02$

i-06f1bd3f330c4c2e0 (HOMEWORK)
PublicIPs: 44.222.61.61 PrivateIPs: 172.31.68.217
CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
```

Chrome File Edit View History Bookmarks Profiles Tab Window Help

OverTheWire: Level 6 | us-east-1 | EC2 Instance Connect | Krypton_Scripts/fre... | No content provided | Vigenere Cipher Sol... | Gemini

dcode.fr/vigenere-cipher

Search for a tool

SEARCH A TOOL ON dCode

e.g. type "boolean"

BROWSE THE FULL dCODE TOOLS' LIST

Results

EICTDGVIZKT_KRN

ABCEDEFGHIJKLMNOPQRSTUVWXYZ (26)

LFSRISNOTRANDOM

audible

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Vigenere Cipher - dCode

Tag(s) : Poly-Alphabetic Cipher

dCode is preparing a new interface. Come test and give your feedback on the new page: **Vigenere Cipher!**

VIGENERE CIPHER

Cryptography - Poly-Alphabetic Cipher - Vigenere Cipher

Summary

- Vigenere Decoder
- Vigenere Encoder
- What is the Vigenere cipher? (Definition)
- How to encrypt using Vigenere cipher?
- How to decrypt Vigenere cipher?
- How to recognize Vigenere ciphertext?
- How to break Vigenere without knowing the key?
- How to find the key when having both cipher and plaintext?
- What are the variants of the Vigenere cipher?
- How to choose the encryption key?
- What is the running key vigenere cipher?
- What is the keyed vigenere cipher?
- What are the advantages of the Vigenere cipher versus Caesar Cipher?
- What is a Saint-Cyr slide?

VIGENERE DECODER

VIGENERE CIPHERTEXT

PRUKLYLRQKQKBE

PLAINTEXT LANGUAGE: English

ALPHABET: ABCDEFGHIJKLMNOPQRSTUVWXYZ

AUTOMATIC DECRYPTION

DECRYPTION METHOD

☒ KNOWING THE KEY/PASSWORD: EICTDGVIZKT_KRN

☐ KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6

☐ KNOWING ONLY A PARTIAL KEY (JOKER+?): FREKEY

☐ KNOWING A PLAINTEXT WORD: CODE

☐ VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

☐ SHOW VIGENERE'S SQUARE/GRID (TABULA RECTA)

DECRYPT

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6

Final Assessment: OTW Krypton Challenge

For Level 6 to 7, I used `cd /krypton/krypton6` and read the *README*, *HINT1*, and *HINT2* files to understand the task. After checking *krypton7*, I created a temporary working directory with `mktemp -d` and moved into it. I made a symbolic link to the key file using `ln -s /krypton/krypton6/keyfile.dat` and set the directory permissions with `chmod 777`. To understand how the encryption program worked, I created test files—like *tale.txt* and a larger file of repeated “A”s—and ran them through `./encrypt6` to see how plaintext was transformed. Using Dcode’s Vigenère tool, I analyzed the ciphertext patterns and compared them to my test inputs. This allowed me to figure out the encryption scheme and successfully decode the *krypton7* file to obtain the password for the next level which is **LFSRISNOTRANDOM**